

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech

Date: 5-03-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Shaik Mohammad Galeeb

Reg. no.: 23BCE9397

Question 1: Student Notes Manager (CRUD Operations)

Develop a simple web application that allows students to create, view, update, and delete study notes stored in MongoDB.

Modules

1. Add Note
2. View Notes
3. Edit Note
4. Delete Note

Functional Requirements

1. Add Note

Student enters:

- Title
- Subject
- Description

AJAX sends JSON data to server.

Example JSON request:

```
{
  "title": "MongoDB Basics",
  "subject": "Database",
  "description": "Introduction to MongoDB concepts"
}
```

MongoDB stores document in **notes collection**.

Example MongoDB document:

```
{
  "_id": "ObjectId",
  "title": "MongoDB Basics",
  "subject": "Database",
  "description": "Introduction to MongoDB concepts",
  "created_date": "2026-03-05"
}
```

2. View Notes

Display all notes stored in database.

AJAX request:

GET /notes

Server returns JSON:

```
[
  {
    "title": "MongoDB Basics",
    "subject": "Database",
```

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```
"description": "Introduction to MongoDB concepts"
}
```

Frontend dynamically renders notes using JavaScript.

3. Update Note

Student can modify title or description.

AJAX request:

PUT /notes/{id}

```
{
  "title": "MongoDB Advanced",
  "description": "Aggregation and indexing"
}
```

MongoDB operation:

```
db.notes.updateOne(
  {_id: ObjectId(id)},
  {$set: {title: "MongoDB Advanced", description: "Aggregation and indexing"}}
)
```

4. Delete Note

AJAX request:

DELETE /notes/{id}

MongoDB query:

```
db.notes.deleteOne({_id: ObjectId(id)})
```

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```
mongosh mongodb://127.0.0.1:27017/
Microsoft Windows [Version 10.0.26200.7922]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell>mongosh
Current Mongosh Log ID: 69a9a708643986fd127c2906
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB: 8.2.5
Using Mongosh: 2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2026-03-05T19:09:45.618+05:30: Access control is not enabled for the database. Read and write access to data and configuration is
unrestricted
-----

test> use student_notes
switched to db student_notes
student_notes> db.createCollection("notes")
{ ok: 1 }
student_notes> db.notes.insertOne({
  title: "MongoDB Basics",
  subject: "Database",
  description: "Introduction to MongoDB concepts",
  created_date: new Date()
})
{
  acknowledged: true,
  insertedId: ObjectId('69a9a799643986fd127c2907')
}
student_notes> db.notes.find()
[
  {
    _id: ObjectId('69a9a799643986fd127c2907'),
    title: 'MongoDB Basics',
    subject: 'Database',
    description: 'Introduction to MongoDB concepts',
    created_date: ISODate('2026-03-05T15:56:09.806Z')
  }
]
student_notes> db.notes.insertMany([
  {
    title: "Machine Learning",
    subject: "AI",
    description: "ML concepts",
    created_date: new Date()
  },
  {
    title: "Data Structures",
    subject: "Programming",
    description: "Trees and graphs",
    created_date: new Date()
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a9a7b1643986fd127c2908'),
    '1': ObjectId('69a9a7b1643986fd127c2909')
  }
}
student_notes> db.notes.find()
```

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```
mongosh mongodb://127.0.0.1 x + v
}
}
student_notes> db.notes.find()
[
  {
    _id: ObjectId('69a9a799643986fd127c2907'),
    title: 'MongoDB Basics',
    subject: 'Database',
    description: 'Introduction to MongoDB concepts',
    created_date: ISODate('2026-03-05T15:56:09.806Z')
  },
  {
    _id: ObjectId('69a9a7b1643986fd127c2908'),
    title: 'Machine Learning',
    subject: 'AI',
    description: 'ML concepts',
    created_date: ISODate('2026-03-05T15:56:33.663Z')
  },
  {
    _id: ObjectId('69a9a7b1643986fd127c2909'),
    title: 'Data Structures',
    subject: 'Programming',
    description: 'Trees and graphs',
    created_date: ISODate('2026-03-05T15:56:33.663Z')
  }
]
student_notes> db.notes.updateOne(
  {title: "MongoDB Basics"},
  {$set: {description: "MongoDB CRUD operations"}}
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
student_notes> db.notes.deleteOne({title: "Machine Learning"})
{ acknowledged: true, deletedCount: 1 }
```

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Question 2: Online Book Finder (Retrieval & Query Operations)

Develop a system where users can search, filter, and retrieve books using different MongoDB queries.

Modules

1. Search Books by Title
2. Filter Books by Category
3. Sort Books by Price or Rating
4. Display Top Rated Books
5. Pagination (Load More)

Database Structure

Collection: **books**

Example document:

```
{
  "_id": "ObjectId",
  "title": "JavaScript Essentials",
  "author": "John Smith",
  "category": "Programming",
  "price": 450,
  "rating": 4.5,
  "year": 2023
}
```

Functional Requirements

1. Search Books by Title

AJAX request:

GET /books/search?title=javascript

MongoDB query:

```
db.books.find({title:{$regex:"javascript",$options:"i"}})
```

Returns JSON results.

2. Filter Books by Category

AJAX request:

GET /books/category/programming

MongoDB query:

```
db.books.find({category:"Programming"})
```

3. Sort Books

Sort by price.

AJAX request:

GET /books/sort/price

MongoDB query:

```
db.books.find().sort({price:1})
```

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Sort by rating.

```
db.books.find().sort({rating:-1})
```

4. Top Rated Books

AJAX request:

GET /books/top

MongoDB query:

```
db.books.find({rating:{$gte:4}}).limit(5)
```

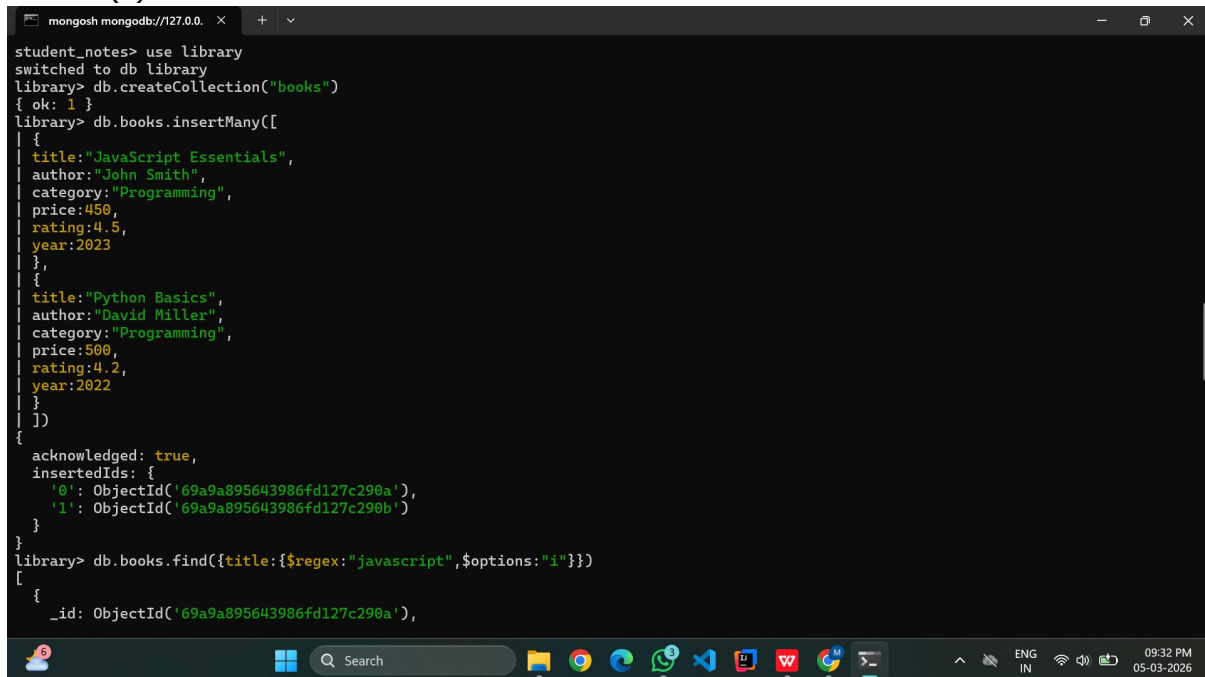
5. Pagination

AJAX request:

GET /books?page=2

MongoDB query:

```
db.books.find()  
  .skip(5)  
  .limit(5)
```



The screenshot shows a MongoDB terminal window with the following commands and output:

```
student_notes> use library  
switched to db library  
library> db.createCollection("books")  
{ ok: 1 }  
library> db.books.insertMany([  
  {  
    title: "JavaScript Essentials",  
    author: "John Smith",  
    category: "Programming",  
    price: 450,  
    rating: 4.5,  
    year: 2023  
  },  
  {  
    title: "Python Basics",  
    author: "David Miller",  
    category: "Programming",  
    price: 500,  
    rating: 4.2,  
    year: 2022  
  }  
)  
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('69a9a895643986fd127c290a'),  
    '1': ObjectId('69a9a895643986fd127c290b')  
  }  
}  
library> db.books.find({title:{$regex:"javascript",$options:"i"}})  
[  
  {  
    _id: ObjectId('69a9a895643986fd127c290a'),
```

The terminal window has a title bar that reads "mongosh mongodb://127.0.0.1:27020". The Windows taskbar at the bottom shows the time as 09:32 PM on 05-03-2026.

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```
mongosh mongodb://127.0.0.1:27000
> use library
library> db.books.find({title:{$regex:"javascript",$options:"i"}})
[
  {
    _id: ObjectId('69a9a895643986fd127c290a'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  }
]
library> db.books.find({category:"Programming"})
[
  {
    _id: ObjectId('69a9a895643986fd127c290a'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a9a895643986fd127c290b'),
    title: 'Python Basics',
    author: 'David Miller',
    category: 'Programming',
    price: 500,
    rating: 4.2,
    year: 2022
  }
]
library> db.books.find().sort({price:1})
[
  {
    _id: ObjectId('69a9a895643986fd127c290a'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a9a895643986fd127c290b'),
    title: 'Python Basics',
    author: 'David Miller',
    category: 'Programming',
    price: 500,
    rating: 4.2,
    year: 2022
  }
]
library> db.books.find({rating:{$gte:4}}).limit(5)
[
  {
    _id: ObjectId('69a9a895643986fd127c290a'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  }
]
```

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```
mongosh mongodb://127.0.0.1:27020
> use library
library> db.books.find({rating:{$gte:4}}).limit(5)
[
  {
    _id: ObjectId('69a9a895643986fd127c290a'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a9a895643986fd127c290b'),
    title: 'Python Basics',
    author: 'David Miller',
    category: 'Programming',
    price: 500,
    rating: 4.2,
    year: 2022
  }
]
library> db.books.find().skip(5).limit(5)
library> db.books.find().skip(5).limit(5)
library> |
```